

Snap Up - Proof of Concept (POC) Requirements

Prepared for Astra Dyne Global Intern Development Team

Following the discussion with the supermarket's Chief Technical Officer (CTO), the following technical requirements must be achieved before proceeding with the Proof of Concept (POC). These requirements are critical for obtaining approval for the pilot implementation.

1. Controlled Database Connectivity Based on Store Presence

Snap Up must establish communication with the supermarket database **only when the customer is physically inside the store**. Once the customer exits the store, all active sessions, API connections, authentication tokens, and database access must be terminated immediately. Even if the customer remains just outside the store premises, the application must **not reconnect** to the supermarket database. The supermarket has proposed using their in-store Wi-Fi network for entry and exit validation during the POC.

Open Technical Discussion: The development team should research and propose alternative methods to satisfy this requirement without depending solely on the supermarket's Wi-Fi. GPS/Latitude-Longitude based validation is **not recommended** due to accuracy limitations. Any secure alternative (e.g., Bluetooth Low Energy beacons, geofencing alternatives, local network authentication, QR/NFC entry validation, etc.) should be evaluated and documented.

2. Demonstrate Database Security

The POC must clearly demonstrate that Snap Up cannot copy, export, or retain the supermarket's database. Only the minimum product information required for barcode lookup and customer shopping should be accessible through secure authenticated APIs using encrypted communication and least-privilege access.

3. High-Speed Barcode Retrieval

Barcode scanning should retrieve product information within approximately **2 seconds or less** under normal operating conditions. The team should optimise barcode decoding, API latency, caching strategy, and database queries to consistently meet this benchmark.

4. Database Retrieval Optimisation

Improve the database retrieval architecture using efficient indexing, lightweight API responses, intelligent caching, pagination where required, and scalable search mechanisms to minimise latency and maximise performance.

Development Action Items

- Analyse each CTO requirement and prepare an implementation strategy.
- Research alternative store-presence detection mechanisms without relying on GPS.
- Document proposed architecture changes and security controls.
- Validate every requirement with measurable test results during the POC.
- Ensure the solution is scalable and production-ready.

Objective: Successfully satisfy all CTO requirements and secure approval for a one-week pilot deployment of Snap Up